Comprehensive Pointer Notes: Water Technology

ENGINEERING CHEMISTRY COURSE MODULE

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Scope: Complete syllabus synthesis encompassing water sources, impurities, hardness chemistry, EDTA titration, boiler troubles, lime-soda/zeolite/ion-exchange softening, water quality metrics (DO, Winkler, BOD, COD), and municipal sewage treatment.

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1. Introduction, Engineering Uses & Sources of Water

- **Preservation of Life:** Basic prime necessity alongside air for sustaining terrestrial organisms and domestic activities.
- **Domestic Applications:** Essential for drinking, cooking, bathing, hygiene, and agricultural farming/irrigation.
- **Importance as an Engineering Material:**
  - **Steam Generation in Boilers:** Primary working fluid in thermal and nuclear power plants.
  - **Hydroelectric Generation:** Kinetic and potential energy source for driving hydraulic turbines.
  - **Steam for Engines:** Working medium for steam locomotives, marine propulsion, and prime movers.
  - **Civil Engineering Construction:** Essential for hydration of cement in concrete, mortar curing, and structural fabrication.
  - **Chemical Process Solvent:** Universal solvent and heat-transfer medium (coolant/reflux) across chemical and petrochemical industries.
- **Main Hydrological Sources of Water:**
  - **Rain Water:** Purest natural form physically; absorbs atmospheric gases (CO_2, SO_2, NO_x) during condensation.
  - **River Water:** Surface runoff containing dissolved surface minerals, agricultural fertilizers, and suspended sand/silt.
  - **Spring or Well Water (Groundwater):** Underground percolation creates high dissolved mineral content (Ca, Mg salts) with low suspended turbidity.
  - **Sea Water:** Ultimate sink; highest salinity (~3.5% dissolved salts, predominantly NaCl with magnesium and calcium salts).

2. Classification of Water Impurities

Natural water is never chemically pure (H_2O) and contains impurities grouped into three broad classes:

- **1. Dissolved Impurities:**
  - **Inorganic Salts:** Carbonates, bicarbonates, sulphates, and chlorides of calcium, magnesium, iron, potassium, and aluminium.
  - **Organic Compounds:** Decomposition products of vegetation, amino acids, proteins, and agricultural runoff.
  - **Dissolved Gases:** Dissolved atmospheric oxygen (O_2), carbon dioxide (CO_2), oxides of nitrogen (NO_x), oxides of sulphur (SO_x), and hydrogen sulphide (H_2S).
- **2. Suspended Impurities:**
  - **Inorganic Suspensions:** Finely divided clay, sand, silt, and silica particles.
  - **Organic Suspensions:** Decaying vegetable matter, animal waste, and biological debris.
- **3. Biological (Microbiological) Impurities:**
  - Pathogenic bacteria (e.g., *Vibrio cholerae*, *Salmonella typhi*), fungi, viral pathogens, and microscopic algae.

3. Hardness of Water, Soap Chemistry & Adverse Effects

- **Definitions:**
  - **Hard Water:** Water that does not produce lather easily (or produces very little lather) with soap, forming an insoluble curdy precipitate.
  - **Soft Water:** Water that readily produces an abundant, persistent lather with a small amount of soap.
- **Fundamental Cause of Hardness:** Presence of dissolved bicarbonates, sulphates, chlorides, and nitrates of bivalent metallic cations, primarily **Calcium (Ca^{2+})** and **Magnesium (Mg^{2+})** (and occasionally Fe^{2+}, Mn^{2+}, Al^{3+}).
- **Chemistry of Soap & Lathering Action:**
  - Soaps are sodium or potassium salts of higher fatty acids (e.g., sodium stearate $C_{17}H_{35}COONa$, oleate, palmitate).
  - **Reaction in Soft Water:** Soap hydrolyzes to generate free stearic acid, which interacts with unhydrolyzed sodium stearate to produce foam/lather:

    > $C_{17}H_{35}COONa + H_2O \rightarrow NaOH + C_{17}H_{35}COOH$ (Stearic Acid)
    > Stearic Acid + Sodium Stearate → Formation of Lather

  - **Reaction in Hard Water:** Soluble sodium stearate reacts immediately with Ca^{2+} and Mg^{2+} ions, precipitating insoluble calcium and magnesium stearates (white curd/scum):

    > $2\ C_{17}H_{35}COONa$ (Soluble Soap) + $Ca^{2+} \rightarrow (C_{17}H_{35}COO)_2Ca\downarrow$ (Insoluble White Ppt) + $2\ Na^+$

> $2\ C_{17}H_{35}COONa\ (Soluble\ Soap) + Mg^{2+} \rightarrow (C_{17}H_{35}COO)_2Mg\downarrow\ (Insoluble\ White\ Ppt) + 2\ Na^+$

  - Lather forms only after all Ca^{2+} and Mg^{2+} ions have been fully precipitated, resulting in massive soap wastage.

- **Adverse Impacts of Hard Water:**
  - **Domestic Use:**
    - **Washing & Bathing:** Wastes large quantities of soap; curdy precipitates adhere to skin and hair, causing irritation.
    - **Drinking:** Ingestion irritates the digestive system; prolonged consumption of high-mineral hard water promotes kidney and bladder stone formation.
    - **Cooking:** Dissolved salts elevate the boiling point of water; curdy deposits coat food surfaces, demanding significantly more fuel and cooking time.
  - **Industrial Use:**
    - **Textile Industry:** Precipitates of Ca/Mg soaps stick to fabric yarns, producing dull shades, patchy streaks, and uneven dye absorption.
    - **Paper Industry:** Calcium and magnesium salts interfere with sizing agents, altering paper texture, finish, and tensile strength.
    - **Sugar Refining:** Dissolved sulphates, nitrates, and carbonates retard sucrose crystallization and reduce sugar yield.
    - **Pharmaceutical Industry:** Forms undesirable side-products, precipitates active drug substances, and contaminates chemical formulations.
  - **Steam Generation in Boilers:** Causes four major catastrophic operational troubles: (1) Priming & Foaming, (2) Scale & Sludge Formation, (3) Caustic Embrittlement, and (4) Boiler Corrosion.

4. Classification of Hardness & Degree of Hardness

- **Classification of Hardness:**
  - **1. Temporary Hardness (Carbonate Hardness):**
    - Caused exclusively by dissolved bicarbonates of calcium and magnesium: **$Ca(HCO_3)_2$** and **$Mg(HCO_3)_2$**.
    - Termed "temporary" because it is easily removed by simple boiling, which thermally decomposes soluble bicarbonates into insoluble precipitates:

      > $Ca(HCO_3)_2 + \Delta \rightarrow CaCO_3\downarrow + H_2O + CO_2\uparrow$
      > $Mg(HCO_3)_2 + \Delta \rightarrow Mg(OH)_2\downarrow + 2\ CO_2\uparrow$

      (Note: $Mg(OH)_2$ precipitates rather than $MgCO_3$ because $Mg(OH)_2$ is significantly more insoluble at boiling temperatures).

- **2. Permanent Hardness (Non-Carbonate Hardness):**
  - Caused by dissolved chlorides, sulphates, and nitrates of calcium and magnesium: **$CaCl_2$, $CaSO_4$, $Ca(NO_3)_2$, $MgCl_2$, $MgSO_4$, $Mg(NO_3)_2$**.
  - Cannot be removed by physical boiling; requires chemical softening techniques (lime-soda, zeolite, or ion-exchange).
- **Total Hardness Relationship:**

> **Total Hardness = Temporary Hardness + Permanent Hardness**

- **Degree of Hardness & $CaCO_3$ Equivalents:**
  - Hardness concentrations and reacting chemical quantities are universally expressed in terms of **$CaCO_3$ equivalent**.
  - **Reasons for Selecting $CaCO_3$ as Standard:**
    - **Molecular Weight is Exactly 100:** Molecular weight = 100, valency = 2, making Equivalent Weight = 50. This greatly simplifies arithmetic conversions.
    - **Lowest Solubility:** Calcium carbonate is the most insoluble salt that readily precipitates out during chemical water softening operations.
  - **Standard Mathematical Conversion Formula:**

> **Hardness in terms of $CaCO_3$ Equivalent (mg/L or ppm) =**
> `[ (Mass of Hardness Salt in mg/L) × 100 ] / [ Molecular Weight of Salt ]`
> `= (Mass of Hardness Salt) × [ 50 / (Chemical Equivalent Weight of Salt) ]`

Reference Data: Molecular Weights, Chemical Equivalents & Multiplication Factors

Hardness Constituent / Ion	Chemical Formula	Molar Mass	Chemical Equivalent	Multiplication Factor
Calcium Bicarbonate	$Ca(HCO_3)_2$	162	81	100 / 162 (or 50/81)
Magnesium Bicarbonate	$Mg(HCO_3)_2$	146 (table 142)	73 (table 71)	100 / 146 (or 50/73)
Calcium Sulphate	$CaSO_4$	136	68	100 / 136 (or 50/68)
Magnesium Sulphate	$MgSO_4$	120	60	100 / 120 (or 50/60)
Calcium Chloride	$CaCl_2$	111	55.5	100 / 111 (or 50/55.5)
Magnesium Chloride	$MgCl_2$	95	47.5	100 / 95 (or 50/47.5)
Calcium Nitrate	$Ca(NO_3)_2$	164	82	100 / 164 (or 50/82)

Hardness Constituent / Ion	Chemical Formula	Molar Mass	Chemical Equivalent	Multiplication Factor
Magnesium Nitrate	$Mg(NO_3)_2$	148	74	100 / 148 (or 50/74)
Calcium Carbonate	$CaCO_3$	100	50	100 / 100 = 1.0
Magnesium Carbonate	$MgCO_3$	84	42	100 / 84 (or 50/42)
Carbon Dioxide	CO_2	44	22	100 / 44 (or 50/22)
Bicarbonate Ion	HCO_3^-	61	61	100 / 122 (or 50/61)
Hydroxide Ion	OH^-	17	17	100 / 34
Carbonate Ion	CO_3^{2-}	60	30	100 / 60 (or 50/30)
Ferrous Chloride	$FeCl_2$	127	63.5	100 / 127
Ferrous Sulphate Heptahydrate	$FeSO_4 \cdot 7H_2O$	278	139	100 / 278
Aluminium Sulphate	$Al_2(SO_4)_3$	342	57	100 / 114 (or 300/342)
Sodium Aluminate	$NaAlO_2$	82	82	100 / 164
Hydrogen Ion	H^+	1	1	100 / 2
Hydrochloric Acid	HCl	36.5	36.5	100 / 73

5. Units of Hardness & Interconversion Relationships

Hardness is expressed in five standard quantitative units:

- **1. Parts Per Million (PPM):** Number of parts of $CaCO_3$ equivalent hardness per 10^6 (one million) parts by weight of water.

  > $1\ ppm = 1\ part\ of\ CaCO_3\ equivalent\ in\ 10^6\ parts\ of\ water$

- **2. Milligrams Per Litre (mg/L):** Number of milligrams of $CaCO_3$ equivalent hardness present per litre of water.

  > $Weight\ of\ 1\ L\ water = 1\ kg = 1{,}000\ g = 10^6\ mg$
  > $\therefore\ 1\ mg/L = 1\ mg\ CaCO_3\ eq\ per\ 10^6\ mg\ water = 1\ part\ per\ 10^6\ parts = \mathbf{1\ ppm}$

- **3. Degree Clarke (°Cl):** Number of grains (1/7000 lb) of $CaCO_3$ equivalent hardness per gallon (10 lb) of water; equivalent to parts of $CaCO_3$ per 70,000 parts of water.

> `1 °Cl = 1 grain of CaCO3 eq per imperial gallon of water = 1 part per 70,000 parts`
> `∴ 1 ppm = 0.07 °Cl ↔ 1 °Cl = 14.3 ppm = 14.3 mg/L`

- **4. Degree French (°Fr):** Number of parts of $CaCO_3$ equivalent hardness per 10^5 (one hundred thousand) parts by weight of water.

> `1 °Fr = 1 part of CaCO3 eq per 10^5 parts of water`
> `∴ 1 °Fr = 10 ppm ↔ 1 ppm = 0.1 °Fr ↔ 1 °Cl = 1.43 °Fr`

- **5. Milliequivalents Per Litre (meq/L):** Number of milliequivalents of hardness present per litre of water.

> `1 meq/L = 1 meq of CaCO3 per litre = 10^-3 × 50 g CaCO3 eq/L = 50 mg/L = 50 ppm`
> `∴ 1 meq/L = 50 ppm = 50 mg/L ↔ 1 ppm = 0.02 meq/L`

> **Master Interconversion Formula:**
> **1 ppm = 1 mg/L = 0.07 °Cl = 0.1 °Fr = 0.02 meq/L**
> *Example Checks:* 1 °Cl = 14.3 ppm = 14.3 mg/L = 1.43 °Fr = 0.286 meq/L.

6. Determination of Hardness by EDTA Method (Complexometric Titration)

- **Titration Principle:**
  - Complexometric titration employing **Ethylenediamine tetraacetic acid (EDTA)**, specifically its disodium salt (Na_2EDTA).
  - EDTA functions as a hexadentate chelating ligand that forms stable, 1:1, colourless chelate complexes with Ca^{2+} and Mg^{2+} ions at an alkaline pH of **9.0 – 10.0**.
  - To maintain constant pH (9–10), an ammonia-ammonium chloride buffer (**NH_4Cl + NH_4OH**) is added.
  - **Indicator:** Alcoholic solution of **Eriochrome Black-T (EBT)**, which exists in a free uncomplexed state as a deep blue dye.
- **Mechanism of Colour Change:**
  - When EBT is introduced into buffered hard water at pH 9–10, it loosely complexes with a fraction of Ca^{2+}/Mg^{2+} ions to form an *unstable wine-red complex*:

> `(Ca2+ + Mg2+) [Hard Water] + EBT [Blue] + Buffer (pH 9–10) → [Ca-EBT + Mg-EBT] [Unstable Wine-Red Complex]`

- During titration, added EDTA first complexes all free uncombined Ca^{2+} and Mg^{2+} ions in solution.
- Because the Metal-EDTA stability constant is far greater than the Metal-EBT stability constant, EDTA displaces EBT from the wine-red complex at the equivalence point:

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[Ca-EBT + Mg-EBT] [Wine-Red] + EDTA → [Ca-EDTA + Mg-EDTA] [Stable Colourless Complex] + EBT [Free Blue Indicator]
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- **End Point:** Extremely sharp color change from **Wine-Red to Pure Blue**.

- **Reagent Preparation Protocols:**
  - **Standard Hard Water (SHW):** Dissolve 1.000 g of pure, dry $CaCO_3$ in minimum dilute HCl. Evaporate to dryness to expel excess acid fumes, then dilute the residue with distilled water to exactly 1.000 L.
    Concentration: 1 mL SHW contains 1.0 mg $CaCO_3$ equivalent hardness (1000 ppm, 0.01 M).
  - **EDTA Titrant Solution:** Dissolve 3.72 g of pure disodium EDTA dihydrate and 0.1 mg $MgCl_2$ in distilled water and dilute to 1 L.
  - **Indicator Solution:** Dissolve 0.5 g Eriochrome Black-T in 100 mL ethanol or methanol.
  - **Buffer Solution (pH 9–10):** Dissolve 67.5 g ammonium chloride (NH_4Cl) in 570 mL concentrated ammonium hydroxide (NH_4OH), and dilute with distilled water to 1 L.

- **Three-Step Experimental Procedure:**
  - **Step 1: Standardization of EDTA Solution:**
    - Pipette 50 mL of SHW into a conical flask. Add 10 mL of buffer solution and 4–5 drops of EBT indicator.
    - Titrate against EDTA solution until the wine-red color sharply shifts to clear blue. Let volume consumed = **V_1 mL**.
    - *Calculation:* 50 mL SHW = 50 mg $CaCO_3$ eq = V_1 mL EDTA → **1 mL EDTA = (50 / V_1) mg $CaCO_3$ eq**.
  - **Step 2: Titration of Unknown Hard Water Sample (Total Hardness):**
    - Pipette 50 mL of raw sample water into a conical flask. Add 10 mL buffer and 4–5 drops EBT indicator.
    - Titrate against standardized EDTA to blue endpoint. Let volume consumed = **V_2 mL**.
    - *Calculation:* 50 mL sample = V_2 × (50 / V_1) mg $CaCO_3$ eq.
      ∴ **Total Hardness (ppm or mg/L) = (V_2 / V_1) × 1000 ppm**.
  - **Step 3: Titration of Boiled Water Sample (Permanent Hardness):**
    - Take 50 mL (or 100 mL) of water sample in a beaker. Boil gently until volume is reduced to approximately half (decomposing all bicarbonates to insoluble $CaCO_3$ and $Mg(OH)_2$).
    - Filter through filter paper to remove precipitates, wash residue with distilled water, cool, and dilute the filtrate back to the original 50 mL volume.

- Add 10 mL buffer and EBT indicator; titrate against EDTA to blue endpoint. Let volume consumed = **V_3 mL**.
- ∴ **Permanent Hardness (ppm or mg/L) = (V_3 / V_1) × 1000 ppm**.

- **Calculation of Temporary Hardness:**

> **Temporary Hardness = Total Hardness − Permanent Hardness = [(V_2 − V_3) / V_1] × 1000 ppm**

7. Boiler Troubles: Causes, Mechanisms & Remediation

Continuous evaporation of feed water in industrial boilers progressively concentrates dissolved salts, precipitating minerals and triggering severe operational failure:

Trouble 1: Priming & Foaming (Carry-Over / Wet Steam Formation)

- **Priming:** Phenomenon wherein steam generated rapidly inside the boiler carries droplets of liquid water along with it into steam delivery headers, generating *wet steam*.
  - **Primary Causes:** (1) High concentration of dissolved solids, (2) Exceptionally high steam velocities, (3) Sudden violent boiling, (4) Defective/improper boiler mechanical design, (5) Sudden surges in steam output demand.
  - **Preventive Measures:** Maintaining proper low water levels, using thoroughly softened feed water, installing mechanical steam purifiers/separators, employing well-designed boilers, avoiding rapid steam load fluctuations, and conducting regular blow-down operations.
- **Foaming:** Phenomenon of persistent foam or stable bubble formation on the boiler water surface, which fails to break and gets swept into the steam pipe.
  - **Primary Causes:** High concentrations of finely divided suspended solids and oily/fatty substances that lower surface tension.
  - **Preventive Measures:** Adding anti-foaming chemicals (e.g., castor oil), removing oily contaminants via coagulation with sodium aluminate ($NaAlO_2$), and executing frequent blow-down.
- **Disadvantages of Priming & Foaming:** Water-level gauge glasses give false readings; wet steam wastes sensible heat, lowering boiler thermal efficiency; dissolved salts carried over deposit in superheaters and blades of steam turbines, inducing blade erosion and unbalanced rotor vibration.

Trouble 2: Sludge & Scale Formation

- **Sludge:**
  - **Characteristics:** Loose, slimy, non-adherent precipitate that settles in cooler portions of the boiler, circulation bends, and mud drums.
  - **Forming Salts:** Salts with greater solubility in hot water than cold water, such as **$Mg(OH)_2$, $MgCO_3$, $MgSO_4$, $CaCl_2$**.

- **Removal:** Easily flushed out during regular periodic blow-down operations or removed with wire brushes.

- **Scale:**
  - **Characteristics:** Hard, dense, firmly adhering crust that bakes directly onto inner boiler tubes and furnace crown plates.
  - **Forming Salts:** Salts whose solubility decreases with increasing temperature, such as **$CaSO_4$, $CaCO_3$, $MgSiO_3$, $CaSiO_3$**.
  - **Removal:** Extremely difficult; requires mechanical chipping with chisels/pneumatic hammers, thermal shock treatment, or chemical dissolution using 5–10% inhibited HCl, EDTA, or sodium carbonate boiling.

- **Disadvantages of Sludge & Scale:**
  - **Wastage of Fuel:** Scales have extraordinarily low thermal conductivity (poor heat conductors); boiler metal must be heated to excessive temperatures to transfer heat, causing massive fuel consumption.
  - **Boiler Tube Overheating & Blistering:** The metal plate overheats, weakens, softens, and bulges under steam pressure.
  - **Danger of Catastrophic Explosion:** When an overheated, scale-insulated tube crack develops, water rushes onto incandescent bare metal, instantaneously flashing into massive volumes of high-pressure steam that rupture the boiler.
  - **Lowering of Boiler Efficiency:** Reduced heat transfer coefficient chokes steaming capacity.

Trouble 3: Caustic Embrittlement

- **Definition:** Form of intercrystalline stress-corrosion cracking that manifests along stressed parts of boiler plates (such as riveted seams, bends, and joint rivets) exposed to high alkaline concentrations, leading to sudden brittle fracture and boiler explosion.
- **Chemical & Electrochemical Mechanism:**
  - In lime-soda softening, a slight excess of sodium carbonate (Na_2CO_3) is added. In high-pressure boilers, Na_2CO_3 hydrolyzes into caustic soda:

    > $Na_2CO_3 + H_2O \rightarrow 2\ NaOH + CO_2\uparrow$

  - Dilute alkaline water seeps into microscopic hairline cracks, rivets, and seams through capillary action.
  - As water continuously evaporates into steam, caustic soda concentrates inside the crevices up to 10% or higher.
  - This establishes an electrochemical **Concentration Cell** between stressed crevice iron and unstressed plane surfaces:

    > `(-) Fe at Stressed Joints & Bends | Conc. NaOH || Dil. NaOH | Fe at Unstressed Plane Surface (+)`

- Iron in the stressed crevice becomes **Anodic** and dissolves as sodium ferrate (Na_2FeO_2):

  $$Fe + 2\ NaOH \rightarrow Na_2FeO_2 + H_2\uparrow$$

- At high boiler temperatures, sodium ferrate decomposes, regenerating caustic soda and depositing magnetic iron oxide (Fe_3O_4):

  $$Na_2FeO_2 + H_2O + O_2 \rightarrow 12\ NaOH + 2\ Fe_3O_4$$

- The regenerated NaOH attacks additional iron along grain boundaries, perpetuating intercrystalline brittle cracking.

- **Preventive Measures:**
  - Adding **Sodium Sulphate (Na_2SO_4)**, tannin, or lignin to boiler feed water. Na_2SO_4 precipitates within hairline crevices, physically sealing them from caustic ingress (maintaining specific Na_2SO_4 to NaOH concentration ratios).
  - Replacing sodium carbonate with **Sodium Phosphate (Na_3PO_4)** as the conditioning/softening agent; phosphate buffers water without generating free caustic alkalinity.

Trouble 4: Boiler Corrosion

Decay and thinning of boiler steel plates caused by chemical or electrochemical attack from environmental species:

- **1. Dissolved Oxygen Attack:**
  - Water contains approximately 8 mg/L of dissolved oxygen at ambient room temperature. At high operating temperatures, O_2 aggressively attacks iron:

    $$2\ Fe + 2\ H_2O + O_2 \rightarrow 2\ Fe(OH)_2\downarrow \text{ (Ferrous Hydroxide)}$$
    $$4\ Fe(OH)_2 + O_2 \rightarrow 2\ [Fe_2O_3 \cdot 2H_2O]\downarrow \text{ (Ferric Oxide / Rust)}$$

  - **Chemical Deaeration:** Addition of calculated scavenging reducing agents:

    $$2\ Na_2SO_3 \text{ (Sodium Sulphite)} + O_2 \rightarrow 2\ Na_2SO_4$$
    $$N_2H_4 \text{ (Hydrazine)} + O_2 \rightarrow N_2\uparrow + 2\ H_2O$$ *(Ideal scavenger: produces only harmless nitrogen gas and water without adding dissolved solids!)*
    $$Na_2S \text{ (Sodium Sulphide)} + 2\ O_2 \rightarrow Na_2SO_4$$

  - **Mechanical Deaeration:** Spraying pre-heated water through atomizing nozzles under partial vacuum in a degassing tower.

- **2. Dissolved Carbon Dioxide Attack:**
  - Enters boiler feed water via bicarbonate thermal decomposition:

    $$Ca(HCO_3)_2 \rightarrow CaCO_3 + CO_2\uparrow + H_2O$$

$$Mg(HCO_3)_2 \rightarrow Mg(OH)_2 + 2\ CO_2\uparrow$$

- CO_2 dissolves in boiler condensate forming carbonic acid (H_2CO_3), which induces slow acidic etching:

$$CO_2 + H_2O \rightarrow H_2CO_3$$

- **Removal:** Adding a calculated quantity of ammonium hydroxide (ammonia neutralization):

$$2\ NH_4OH + CO_2 \rightarrow (NH_4)_2CO_3 + H_2O$$

- **3. Acids from Hydrolysis of Dissolved Magnesium Salts:**
  - Magnesium salts (especially $MgCl_2$) hydrolyze under high temperature and pressure, liberating hydrochloric acid:

$$MgCl_2 + 2\ H_2O \rightarrow Mg(OH)_2\downarrow + 2\ HCl$$

  - Liberated HCl reacts with boiler steel in a dangerous, self-propagating cyclic catalytic corrosion loop:

$$Fe + 2\ HCl \rightarrow FeCl_2 + H_2\uparrow$$
$$FeCl_2 + 2\ H_2O \rightarrow Fe(OH)_2 + 2\ HCl \quad \textit{(HCl is regenerated to attack more iron!)}$$
$$Fe(OH)_2 + H_2O + 1/2\ O_2 \rightarrow Fe_2O_3 \cdot 3H_2O \text{ (Ferric Oxide Rust)}$$

  - ∴ Even a minute residual trace of $MgCl_2$ produces catastrophic, continuous metal decay.
  - **Prevention:** Complete removal of $MgCl_2$ via chemical softening; dosing alkaline corrosion inhibitors (sodium silicates, sodium phosphates, sodium chromate); executing frequent blow-down.

8. Industrial Methods for Water Softening

Softening is the process of eliminating hardness-producing multivalent cations from water. The three principal industrial methods are:

Method 1: Lime-Soda Process

- **Fundamental Principle:** Converts soluble calcium and magnesium salts into insoluble precipitates of **Calcium Carbonate ($CaCO_3$)** and **Magnesium Hydroxide ($Mg(OH)_2$)** by adding stoichiometrically calculated quantities of **Hydrated Lime [$Ca(OH)_2$]** and **Soda Ash [Na_2CO_3]**. Free acids, CO_2, and H_2S are neutralized simultaneously.
- **Stoichiometric Reaction Matrix & Reagent Requirements:**

Water Constituent / Hardness Salt	Governing Chemical Reaction	Reagent Required
$Ca(HCO_3)_2$ (Temp Ca)	$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2\ CaCO_3\downarrow + 2\ H_2O$	**L**
$Mg(HCO_3)_2$ (Temp Mg)	$Mg(HCO_3)_2 + 2\ Ca(OH)_2 \rightarrow 2\ CaCO_3\downarrow + Mg(OH)_2\downarrow + 2\ H_2O$	**2 L**
$CaSO_4$, $CaCl_2$ (Perm Ca)	$CaSO_4$ / $CaCl_2 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$ / 2 NaCl	**S**
$MgSO_4$, $MgCl_2$ (Perm Mg)	$MgSO_4$ / $MgCl_2 + Ca(OH)_2 + Na_2CO_3 \rightarrow Mg(OH)_2\downarrow + CaCO_3\downarrow + Na_2SO_4$ / 2 NaCl	**L + S**
Dissolved CO_2	$CO_2 + Ca(OH)_2 \rightarrow CaCO_3\downarrow + H_2O$	**L**
Free Mineral Acids (H^+, HCl, H_2SO_4)	$2\ HCl + Ca(OH)_2 + Na_2CO_3 \rightarrow CaCO_3\downarrow + 2\ NaCl + 2\ H_2O$	**L + S**
Bicarbonate Ion (HCO_3^- / $NaHCO_3$)	$2\ NaHCO_3 + Ca(OH)_2 \rightarrow CaCO_3\downarrow + Na_2CO_3 + 2\ H_2O$	**L – S** *(Saves Soda!)*
Coagulant: $FeSO_4$	$FeSO_4 + Ca(OH)_2 \rightarrow Fe(OH)_2\downarrow + CaSO_4$; $CaSO_4 + Na_2CO_3 \rightarrow CaCO_3\downarrow + Na_2SO_4$	**L + S**
Coagulant: $Al_2(SO_4)_3$	$Al_2(SO_4)_3 + 3\ Ca(OH)_2 \rightarrow 2\ Al(OH)_3\downarrow + 3\ CaSO_4$; $3\ CaSO_4 + 3\ Na_2CO_3 \rightarrow 3\ CaCO_3\downarrow + 3\ Na_2SO_4$	**L + S**
Coagulant: $NaAlO_2$	$NaAlO_2 + 2\ H_2O \rightarrow Al(OH)_3\downarrow + NaOH$; NaOH reacts with CO_2/HCO_3^-	**– L** *(Saves Lime!)*

- **Non-Hardness Causing Substances That DO NOT Consume Lime or Soda:**

> **$NaCl$, KCl, SiO_2, Na_2SO_4, K_2SO_4, Fe_2O_3**

  - **NaCl & KCl:** Monovalent alkali halides; do not impart hardness and do not participate in precipitation reactions with $Ca(OH)_2$ or Na_2CO_3.
  - **SiO_2 (Silica):** Inert suspended or colloidal sand particles; insoluble and non-reactive with lime or soda.
  - **Na_2SO_4 & K_2SO_4:** Soluble, completely neutral alkali sulphates; share common ions with reaction products without consuming reagents.
  - **Fe_2O_3:** Insoluble ferric oxide in an already oxidized state; non-reactive towards softening chemicals.

Mathematical Formulation for Lime & Soda Requirements

Using the chemical equivalence where 100 parts by mass of $CaCO_3$ ≡ 74 parts of $Ca(OH)_2$ ≡ 106 parts of Na_2CO_3:

> **Lime Required (kg or g) =**
> [74 / 100] × [Temp Ca^{2+} + 2 × Temp Mg^{2+} + Perm (Mg^{2+} + Fe^{2+} + Al^{3+}) + CO_2 + H^+ + HCO_3^- − $NaAlO_2$] × [V × 100] / [1000 × (% Purity of Lime)]
>
> **Soda Required (kg or g) =**
> [106 / 100] × [Perm (Ca^{2+} + Mg^{2+} + Fe^{2+} + Al^{3+}) + H^+ − HCO_3^-] × [V × 100] / [1000 × (% Purity of Soda)]
> *(Note: All constituent quantities must be converted into their respective $CaCO_3$ equivalents before substitution; V = volume of water in litres).*

Comparison: Cold vs. Hot Lime-Soda Processes

Operating Feature	Cold Lime-Soda Process	Hot Lime-Soda Process
Operating Temperature	Room Temperature (Ambient, 25°C – 30°C)	Elevated: 80°C – 150°C (near boiling point)
Reaction Kinetics	Slow; requires vigorous continuous mechanical stirring	Very rapid (proceeds many-fold faster)
Coagulant Requirement	Essential (Alum, $FeSO_4$, $NaAlO_2$ needed to coagulate fine ppt)	No coagulants required (precipitates aggregate and settle rapidly)
Dissolved Gas Removal	Gases (CO_2, air) remain dissolved	Thermal deaeration drives out dissolved CO_2, air, and gases
Filtration Ease	Higher water viscosity; wood-fibre / sand filters clog faster	Low viscosity of hot water facilitates rapid sand filtration
Residual Hardness	**50 – 60 ppm**	**15 – 30 ppm**

- **Advantages of Lime-Soda Process:** Economical; substantially reduces total dissolved mineral solids; alkaline pH kills pathogenic bacteria; effectively removes iron and manganese; improves corrosion resistance.
- **Disadvantages of Lime-Soda Process:** Residual hardness (15–60 ppm) is inadequate for modern high-pressure boilers; generates massive quantities of wet sludge requiring disposal (often used for landfilling low-lying city areas); requires precise chemical dosing and skilled supervision.

Method 2: Zeolite (Permutite) Process

- **Chemical Composition:** Hydrated sodium alumino-silicate capable of reversibly exchanging its active sodium cations for hardness-producing bivalent cations (Ca^{2+}, Mg^{2+}):

> **$Na_2O \cdot Al_2O_3 \cdot xSiO_2 \cdot yH_2O$** (where x = 2–10 and y = 2–6; represented as **Na_2Ze**)

- **Types of Zeolites:**
  - **Natural Zeolite:** Non-porous, green-sand minerals; e.g., Natrolite [$Na_2O \cdot Al_2O_3 \cdot 4SiO_2 \cdot 2H_2O$].
  - **Synthetic Zeolite:** Highly porous, gel-structure silicates synthesized by fusing china clay, feldspar, and soda ash; exhibits significantly higher ion-exchange capacity.
- **Softening Reactions:** Hard water percolates through a packed bed of sodium zeolite:

> $Na_2Ze + Ca(HCO_3)_2 \rightarrow CaZe + 2\ NaHCO_3$
> $Na_2Ze + Mg(HCO_3)_2 \rightarrow MgZe + 2\ NaHCO_3$
> $Na_2Ze + CaSO_4\ /\ CaCl_2 \rightarrow CaZe + Na_2SO_4\ /\ 2\ NaCl$
> $Na_2Ze + MgSO_4\ /\ MgCl_2 \rightarrow MgZe + Na_2SO_4\ /\ 2\ NaCl$

- **Regeneration of Exhausted Zeolite:**
  - When active Na^+ sites are fully converted to CaZe and MgZe, the bed is exhausted.
  - Regenerated by soaking and flushing with a **10% Brine (Sodium Chloride, NaCl)** solution:

> $CaZe + 2\ NaCl \rightarrow Na_2Ze$ (Active Bed Regenerated) + $CaCl_2$ (Discharged to Washings/Sink)
> $MgZe + 2\ NaCl \rightarrow Na_2Ze$ (Active Bed Regenerated) + $MgCl_2$ (Discharged to Washings/Sink)

- **Quantitative Stoichiometry:**

> 1 chemical equivalent of NaCl = 1 chemical equivalent of $CaCO_3$
> **58.5 g NaCl ≡ 50.0 g $CaCO_3$ equivalent**

- **Advantages of Zeolite Process:**
  - Produces soft water of very low residual hardness (**~10 ppm**).
  - Compact equipment requiring minimal physical space.
  - Extremely clean; zero precipitate or sludge formation.
  - Bed automatically self-adjusts to incoming fluctuations in raw water hardness.
  - Short softening cycle; simple operation with minimal labor skill.
- **Disadvantages of Zeolite Process:**
  - Treated water has higher sodium salt content than lime-soda water.
  - Does not remove acidic anions (HCO_3^-, CO_3^{2-}); generated $NaHCO_3$ decomposes in boilers to release CO_2 (causing corrosion) and NaOH (causing caustic embrittlement).
  - Turbid water clogs zeolite pores (raw water turbidity must be <10 ppm).
  - Acidic water hydrolyzes the aluminosilicate mineral matrix, destroying the zeolite bed.

- Heavy metals (Fe^{2+}, Mn^{2+}) form irreversible iron/manganese zeolites that cannot be regenerated by brine.

Method 3: Ion-Exchange (Demineralization / Deionization) Process

- **Principle:** Completely removes *all* dissolved mineral cations and anions using insoluble, cross-linked, long-chain synthetic organic polymers (resins) with a microporous matrix, yielding pure demineralized (deionized) water.
- **1. Cation Exchange Resins (CXR):**
  - Styrene-divinylbenzene (DVB) cross-linked copolymers functionalized by sulphonation ($-SO_3H$) or carboxylation ($-COOH$); represented as **RH^+**.
  - **Softening Reactions:** Exchanges acidic H^+ ions for incoming multivalent and monovalent metal cations:

    $$2\ RH^+ + Ca^{2+} \rightarrow R_2Ca + 2\ H^+$$
    $$2\ RH^+ + Mg^{2+} \rightarrow R_2Mg + 2\ H^+$$
    $$RH^+ + Na^+ \rightarrow RNa + H^+$$

  - **Regeneration:** Exhausted resin (R_2Ca, R_2Mg) is regenerated by backwashing with **dilute Hydrochloric Acid (HCl) or Sulphuric Acid (H_2SO_4)**:

    $$R_2Ca + 2\ HCl \rightarrow 2\ RH^+ \text{ (Regenerated Resin)} + CaCl_2 \text{ (Washings to Sink)}$$

- **2. Anion Exchange Resins (AXR):**
  - Styrene-divinylbenzene copolymers functionalized with basic quaternary ammonium ($-N^+R_3OH^-$) or amino groups; represented as **$R'OH^-$**.
  - **Softening Reactions:** Exchanges basic OH^- ions for incoming mineral anions:

    $$R'OH + Cl^- \rightarrow R'Cl + OH^-$$
    $$2\ R'OH + SO_4^{2-} \rightarrow R'_2SO_4 + 2\ OH^-$$
    $$R'OH + HCO_3^- \rightarrow R'HCO_3 + OH^-$$

  - **Regeneration:** Exhausted resin is regenerated by passing **dilute Caustic Soda (NaOH)** solution:

    $$R'_2SO_4 + 2\ NaOH \rightarrow 2\ R'OH \text{ (Regenerated Resin)} + Na_2SO_4 \text{ (Washings to Sink)}$$

- **Neutralization & Degasification:**
  - H^+ released from CXR combines with OH^- released from AXR to produce pure neutral water:

    $$H^+ + OH^- \rightarrow H_2O$$

  - Water exiting the CXR tower contains carbonic acid (H_2CO_3), which is passed through a **Degasifier / Decarbonation Tower** where countercurrent air strips out CO_2:

$$H_2CO_3 \rightarrow H_2O + CO_2\uparrow$$

- **Advantages of Demineralization:** Produces water with extremely low residual hardness (**~2 ppm**); can treat highly acidic or alkaline waters; ideal boiler feed water for modern high-pressure steam power plants.
- **Disadvantages of Demineralization:** High initial capital cost of resin columns; regenerant chemicals (HCl, NaOH) are expensive and corrosive; raw water turbidity must be <10 ppm to prevent mechanical fouling of resin pores.

9. Water Quality Analysis (DO, Winkler Titration, BOD, COD)

1. Dissolved Oxygen (DO) Dynamics

- **Definition:** Amount of free, non-compound gaseous oxygen (O_2) dissolved in an aqueous body.
- **Critical Threshold:** Minimum concentration of **4 – 5 mg/L** is strictly required for the survival and propagation of healthy aquatic fish and aerobic biota.
- **Natural Aeration Mechanisms:** Atmospheric diffusion (surface turbulence and wind agitation) and aquatic photosynthesis by submerged flora/algae.
- **Physicochemical Factors Controlling Oxygen Solubility:**
  - **Water Temperature:** Inversely proportional; cold water holds significantly higher dissolved oxygen, whereas elevated temperatures lower gas solubility.
  - **Salinity Concentration:** Freshwater holds substantially more dissolved oxygen than brackish or marine sea water.
  - **Organic Load:** Heavy organic pollution triggers rapid bacterial proliferation, depleting dissolved oxygen through aerobic respiration.

2. The Winkler Titration Method for Dissolved Oxygen

- **Overview:** Classic iodometric titration technique wherein unstable dissolved oxygen is chemically "fixed" into a stable manganese precipitate to prevent atmospheric escape before quantification.
- **Four-Step Analytical Procedure & Chemistry:**
  - **Step 1: Sample Fixing:** Collect sample without aeration in a glass-stoppered BOD bottle. Add manganese sulphate ($MnSO_4$) and alkaline potassium iodide reagent (NaOH + KI + NaN_3). A precipitate forms:

    $$MnSO_4 + 2\ NaOH \rightarrow Mn(OH)_2\ (White\ Floc) + Na_2SO_4$$
    $$2\ Mn(OH)_2 + O_2\ (Dissolved) \rightarrow 2\ MnO(OH)_2\downarrow\ (Basic\ Manganic\ Hydroxide\ /\ Brown\ Floc)$$

    (Presence of brown precipitate confirms dissolved oxygen; white precipitate indicates zero DO).
  - **Step 2: Acidification:** Add concentrated sulphuric acid (H_2SO_4). The brown floc dissolves, oxidizing iodide ions to free molecular iodine:

> $MnO(OH)_2 + 2\ H_2SO_4 + 2\ KI \rightarrow MnSO_4 + K_2SO_4 + 3\ H_2O + I_2$ (Liberated Free Iodine)

- **Step 3: Thiosulphate Titration:** Titrate the liberated free iodine against standard sodium thiosulphate ($Na_2S_2O_3$, typically 0.025 N) until dark golden yellow shifts to pale straw yellow:

> $I_2 + 2\ Na_2S_2O_3 \rightarrow 2\ NaI + Na_2S_4O_6$ (Sodium Tetrathionate)

- **Step 4: Starch Endpoint:** Add 1 mL fresh starch indicator solution (forming a deep blue starch-iodine adsorption complex). Continue titrating dropwise until the intense blue turns completely **clear and colourless**.

- **Stoichiometric Stopper Ratio:**

> **1 mole of Dissolved O_2 ≡ 2 moles of Liberated I_2 ≡ 4 equivalents of $Na_2S_2O_3$**

3. Biochemical Oxygen Demand (BOD_5)

- **Definition:** The quantity of dissolved free oxygen consumed by aerobic microorganisms to biologically oxidize decomposable organic matter present in sewage or wastewater under standard incubation conditions at **20°C for 5 days**.

> $Organic\ Matter + O_2 \rightarrow CO_2 + H_2O$

- **Standard Normative Values:**
  - Average Municipal Sewage: **100 – 150 ppm**.
  - Potable Drinking Water: **< 1 ppm**.

- **Analytical Determination Protocol:**
  - A known volume of sample is diluted with nutrient-rich aerated distilled water (ensuring sufficient oxygen is available for full 5-day aerobic oxidation).
  - Diluted sample is split into two airtight BOD bottles:
    - **Bottle 1 (Initial / Blank DO, DO_b):** Dissolved oxygen determined immediately via Winkler titration.
    - **Bottle 2 (Incubated DO, DO_i):** Incubated in the dark at 20°C for exactly 5 days; residual DO determined via Winkler titration.
  - **Calculation Formula:**

> **BOD_5 (mg/L or ppm) = (DO_b − DO_i) × Dilution Factor**
> where Dilution Factor = [Volume of sample after dilution] / [Volume of sample before dilution]

- **Environmental Engineering Significance:**
  - Directly quantifies the biodegradable organic pollution load of wastewater.
  - Measures the removal efficiency of biological wastewater treatment plants.

- Determines the natural self-purification capacity of receiving rivers and streams (modeling Streeter-Phelps dissolved oxygen sag curves).

4. Chemical Oxygen Demand (COD)

- **Definition:** The amount of oxygen required for the complete chemical oxidation of both biodegradable and non-biodegradable organic matter in wastewater under boiling acidic conditions using a powerful chemical oxidizing agent (**acidified Potassium Dichromate, $K_2Cr_2O_7$**) over **2 to 3 hours**.

> $C_xH_yO_z + (x + y/4 - z/2)\ O_2 \rightarrow x\ CO_2 + (y/2)\ H_2O$

- **Analytical Determination Protocol:**
  - A known volume of wastewater is refluxed for 2–3 hours with a known excess of standard 0.25 N $K_2Cr_2O_7$ and concentrated H_2SO_4.
  - **Silver Sulphate (Ag_2SO_4) Catalyst:** Added to accelerate and facilitate the oxidation of refractory straight-chain aliphatics and aromatic compounds.
  - **Mercuric Sulphate ($HgSO_4$) Masking Agent:** Added to complex and precipitate chloride ions (as $HgCl_2$), preventing chloride oxidation by dichromate which would falsely elevate COD values.
  - Organic carbon is completely oxidized to CO_2 and H_2O. The excess unreacted dichromate is back-titrated against standard **Ferrous Ammonium Sulphate (FAS / Mohr's Salt: $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$)** using **Ferroin Indicator** (1,10-phenanthroline ferrous complex; endpoint from bluish-green to wine-red):

    > $Cr_2O_7^{2-} + 14\ H^+ + 6\ Fe^{2+} \rightarrow 2\ Cr^{3+} + 7\ H_2O + 6\ Fe^{3+}$

  - A parallel blank titration is performed using distilled water instead of wastewater.
  - **Calculation Formula:**

    > **$COD\ (mg/L\ or\ ppm) = [\ (V_1 - V_2) \times N_{FAS} \times 8000\ ] / [\ Sample\ Volume\ (mL)\ ]$**
    > where:
    > V_1 = Volume of FAS consumed in blank titration (mL)
    > V_2 = Volume of FAS consumed in sample back-titration (mL)
    > N_{FAS} = Normality of standard FAS solution
    > 8000 = Equivalent weight of oxygen (8 g/eq) × 1000 mg/g conversion factor

- **Engineering Significance:** Rapid total turnaround (2–3 hours vs. 5 days for BOD); completely unaffected by industrial toxins, bactericides, or environmental conditions hostile to micro-organisms; universally utilized for monitoring industrial discharge permits and designing wastewater plant aeration basins.

Comparative Matrix: BOD vs. COD

Comparative Feature	Biochemical Oxygen Demand (BOD)	Chemical Oxygen Demand (COD)
Oxidation Mechanism	Biological oxidation via living aerobic micro-organisms	Chemical oxidation via acidified Potassium Dichromate ($K_2Cr_2O_7$)
Testing Duration	5 Days (incubated strictly at 20°C in the dark)	2 to 3 Hours (under boiling reflux)
Organic Matter Scope	Measures **Biodegradable** organic matter only	Measures **Total** organic matter (Biodegradable + Chemically Oxidizable Non-Biodegradable)
Relative Quantitative Value	Always lower than COD (**BOD < COD**)	Always higher than BOD (**COD > BOD**)
Effect of Toxic Pollutants	Severely inhibited by toxins, heavy metals, extreme pH, or bactericides	Unaffected by bactericidal toxins or microbial poisons
Standard Catalyst Used	Nutrient buffer seeding	Silver sulphate (Ag_2SO_4) + Mercuric sulphate ($HgSO_4$)

10. Sewage Treatment & Wastewater Engineering

- **Definition of Sewage:** Complex liquid waste stream combining domestic wastewater, human excreta, industrial wash waters, groundwater infiltration, and street stormwater runoff.
- **Primary Objectives of Treatment:** Prevent contamination of natural receiving water bodies, eliminate foul and septic odors, eliminate human disease transmission, and prevent the destruction of fish and aquatic ecosystems.
- **Four-Stage Municipal Wastewater Treatment Workflow:**
  - **Stage 1: Preliminary Treatment (Screening & Grit Removal):**
    - **Objective:** Removal of coarse, floating, and large suspended inorganic/organic debris (wood fragments, cloth rags, paper, plastic, gravel, oil/grease).
    - **Unit Operations:** Bar screens, coarse/fine mesh screens, and grit chambers to protect downstream pumps from abrasion.
  - **Stage 2: Primary Treatment (Sedimentation with Coagulation):**
    - **Objective:** Economical gravity separation of finely divided suspended organic solids and colloidal turbidity.
    - **Process:** Wastewater enters quiescent sedimentation tanks where chemicals (alum [$Al_2(SO_4)_3 \cdot 18H_2O$], hydrated lime) are added. These form heavy gelatinous flocs that entrap colloidal particles, settling them out as primary sludge.
  - **Stage 3: Secondary (Biological) Treatment:**

- **Objective:** Aerobic biological oxidation of dissolved and colloidal organic matter into stable, harmless mineralized end-products (CO_2, H_2O, NH_3, NO_2^-, NO_3^-).
- **Key Industrial Processes:**
  - **(a) Trickling (Sprinkling) Filters:**
    - Effluent is sprayed via rotating overhead distributors over a bed of crushed basalt stone or gravel (1–3 m deep).
    - As wastewater trickles downward over media surfaces, a gelatinous microbial zoogleal film (aerobic bacteria, protozoa, fungi) absorbs organic waste and oxidizes it aerobically. Treated effluent exits at the bottom.
  - **(b) Activated Sludge Process:**
    - Primary sedimented effluent is mixed with a calculated fraction of recycled *activated sludge* (sludge densely populated with active, acclimated aerobic micro-organisms).
    - The mixture (mixed liquor) is vigorously aerated and agitated in aeration tanks for **4 to 6 hours** via compressed air diffusers or mechanical surface aerators.
    - Effluent flows into a secondary clarifier where flocculated microbial biomass rapidly settles as activated sludge.
    - A portion of settled sludge is recycled back to inoculate incoming feed, while surplus sludge is digested anaerobically to produce biogas (methane) or dewatered for agricultural fertilizer.

- **Stage 4: Tertiary (Advanced Polishing) Treatment:**
  - **Objective:** Final polishing to eliminate residual nutrients (phosphorus, nitrogen), dissolved organics, and microbial pathogens prior to environmental discharge or water reclamation.
  - **Key Unit Processes:**
    - **Phosphorus Precipitation:** Dosing with lime [$Ca(OH)_2$], alum, or ferric chloride to precipitate soluble orthophosphates as insoluble calcium hydroxyapatite / calcium phosphate:

      $$3\ Ca(OH)_2 + 2\ PO_4^{3-} \rightarrow Ca_3(PO_4)_2\downarrow + 6\ OH^-$$

    - **Nitrogen Stripping:** Raising pH to >10.5 using lime, followed by countercurrent air stripping in packed towers to drive out dissolved ammonia gas ($NH_3\uparrow$).
    - **Disinfection:** Final chlorination (or UV radiation / ozonation) to kill remaining pathogenic bacteria and viruses before discharging treated water into rivers, lakes, or coastal waters.

11. Key Formula Summary for Engineering Calculations

Calculation Metric	Mathematical Formula & Core Relationships
$CaCO_3$ Equivalent Hardness	`[ (Mass of Salt in mg/L) × 100 ] / [ Molecular Weight of Salt ]`

Calculation Metric	Mathematical Formula & Core Relationships
Hardness Unit Equivalence	1 ppm = 1 mg/L = 0.07 °Cl = 0.1 °Fr = 0.02 meq/L 1 °Cl = 14.3 ppm; 1 °Fr = 10 ppm; 1 meq/L = 50 ppm
EDTA Method Standardization	1 mL of EDTA = [50 / V_1] mg of $CaCO_3$ eq *(where V_1 = mL EDTA for 50 mL SHW)*
Total Hardness (EDTA)	Total Hardness (ppm) = [V_2 / V_1] × 1000 *(for 50 mL water sample)*
Permanent Hardness (EDTA)	Permanent Hardness (ppm) = [V_3 / V_1] × 1000 *(for 50 mL boiled sample)*
Temporary Hardness (EDTA)	Temporary Hardness = Total Hardness − Permanent Hardness = [(V_2 − V_3) / V_1] × 1000 ppm
Lime Requirement (kg)	[74 / 100] × Σ($CaCO_3$ eq of L-consumers) × [V × 100] / [1000 × % Purity]
Soda Requirement (kg)	[106 / 100] × Σ($CaCO_3$ eq of S-consumers) × [V × 100] / [1000 × % Purity]
Zeolite Softener Regeneration	1 eq NaCl = 1 eq $CaCO_3$ → 58.5 g NaCl ≡ 50 g $CaCO_3$ eq Hardness (ppm) = [(Mass of NaCl in g × 50 / 58.5) × 1000] / [Volume of Water (L)]
Ion-Exchange Regeneration	1 L of 1 N Acid/Base = 50 g of $CaCO_3$ equivalent
BOD_5 (mg/L)	BOD_5 = (DO_b − DO_i) × [Volume after dilution / Volume before dilution]
COD (mg/L)	COD = [(V_1 − V_2) × N_{FAS} × 8000] / [Sample Volume (mL)]